value when aggregated, charted, or further analyzed. Regardless of your favorite way to collect data, you should use some best practices when you do logging to ensure that the data you're saving is something you can work with. After all, if you're not doing that, what's the point of writing logs in the first place?

Use timestamps, specifically in ISO 8601

Anything you log should have a timestamp. Most logging frameworks take care of this for you, but to help make your logs even more searchable and readable use the standard date format ISO 8601 (en.wikipedia.org/wiki/ISO_8601). ISO 8601 is a date/time format that's human readable, includes time zone information, and best of all is a standard that's supported in most development languages. The following code shows the difference between an ISO 8601 date and a UNIX epoch timestamp, which unfortunately many people still use.



Which would you rather read? The most annoying thing about the UNIX format is that if you are at all concerned with time zones you need an additional offset to figure out when the date happened relative to UTC time.

Use unique IDs in your logs, but be careful of consumer data

If you want to be able to track down specific events in your logs, you should create IDs that you can index and search on to track transactions through complex systems. But if you do this, make sure you're not using any kind of sensitive data that could enable hackers to find out more about consumers, like their Social Security number or other personal information.

Use standard log categories and frameworks

Log4J and other Log4 frameworks make adding logging to your application a breeze. Instead of trying to reinvent the wheel, use these common